

What the Stokes stream function means

A note on the initial field in the axisymmetric setup. 4 October 2026.

The short version

The Stokes stream function is a way to draw an incompressible flow in a slice through a round pipe or a vortex, so that volume is conserved automatically. It is not the Stokes equations, and it is not a solution of Navier-Stokes. It only builds the starting velocity.

Why it exists

In cylindrical coordinates (r, θ, z) , an axisymmetric field does not depend on the angle θ . Continuity, the statement that fluid is not created or destroyed, is then one equation for two meridional components, u_r and u_z . A single function $\psi(r, z)$ can carry both:

$$u_r = -(1/r) d(\psi)/dz, u_z = (1/r) d(\psi)/dr$$

Fluid moves along the curves of constant ψ . None crosses from one curve to another. The swirl, u_θ , does not enter these two formulas. Continuity does not fix it, so it can be added separately.

The formula in the report

The report sets $\psi_0 = r^2 \text{ times } z$, times a Gaussian bump centered at $r = r_0$. Differentiating that gives the written velocity:

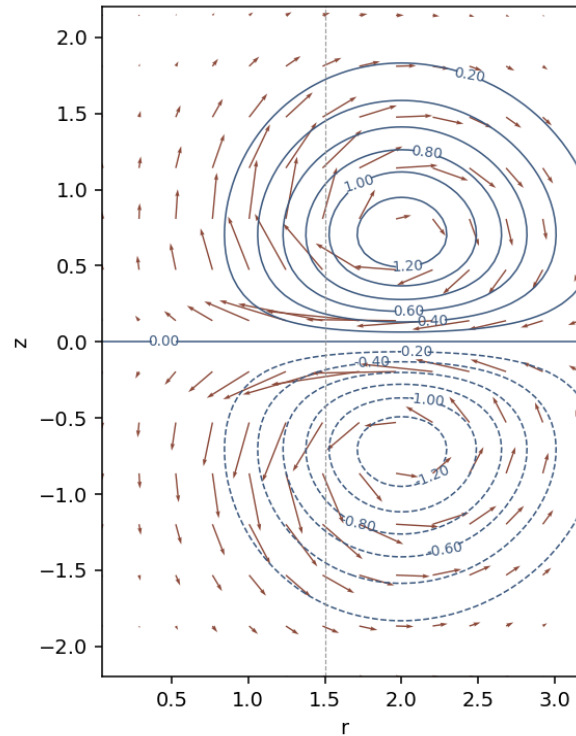
$$u_r = -r (1 - 2 z^2 / \alpha^2) \phi$$

$$u_z = z (2 - 2 r (r - r_0) / \alpha^2) \phi$$

ϕ is the same exponential. A numerical derivative of ψ matches these expressions to about 10^{-11} . The factor r^2 in ψ is what keeps u_r from blowing up on the axis: near $r = 0$, u_r behaves like r , which is allowed. The line $u_\theta = r \phi$ is swirl added by hand. It is not produced by ψ .

What the figure shows

Stokes stream function (lines) and meridional velocity (arrows)



Blue curves are constant ψ . Brown arrows are the meridional velocity (u_r, u_z). The arrows follow the curves. The dashed line is $r = r_0$, the center of the Gaussian. Above $z = 0$ the circulation is one way; below $z = 0$ it reverses, because ψ itself changes sign with z . The zero contour on $z = 0$ is the dividing line between those two cells.

What this does and does not mean

It means the starting field is smooth, decays away from the bump, and is divergence-free in the r - z plane. That is the whole job of the stream function. It does not step the flow forward, it does not include pressure, and it does not produce a finite-time singularity. The discrete script does not use this field. It starts from flat velocities on a periodic cube, and it does not enforce continuity.

Checked against a central difference of ψ at $r = 1.2, z = 0.4$, with $r_0 = 1.5$ and $\alpha = 1$. The analytic u_r and u_z agree to about $1e-11$.